

60 Short Essay Questions – Principles of Radiotherapy

- 1. What is radiotherapy?*
- 2. What is the main goal of radiotherapy?*
- 3. What types of radiation are commonly used in radiotherapy?*
- 4. What is external beam radiotherapy?*
- 5. What is brachytherapy?*
- 6. What is the role of a linear accelerator in radiotherapy?*
- 7. Why is accurate patient positioning important in radiotherapy?*
- 8. What is fractionation in radiotherapy?*
- 9. Why is fractionation beneficial?*
- 10. What is a target volume?*
- 11. What is the difference between tumor and normal tissue response?*
- 12. What is dose in radiotherapy?*

13. What unit is used to measure absorbed dose?

14. What is radiosensitivity?

15. What factors affect radiosensitivity?

16. What is oxygen effect in radiotherapy?

17. What is the therapeutic ratio?

18. What is treatment planning?

19. What is simulation in radiotherapy?

20. What imaging modalities are commonly used in radiotherapy planning?

21. What is conformal radiotherapy?

22. What is intensity-modulated radiotherapy (IMRT)?

23. What is image-guided radiotherapy (IGRT)?

24. What is stereotactic radiotherapy?

25. What is the purpose of shielding in radiotherapy?

26. What is beam collimation?

27. What is a multileaf collimator (MLC)?

28. What is the biological effect of radiation on cells?

29. What is acute radiation reaction?

30. What is late radiation reaction?

31. What is a dose-limiting organ?

32. What is a tolerance dose?

33. What is treatment verification?

34. What is portal imaging?

35. What is radiation protection in radiotherapy?

36. What is quality assurance in radiotherapy?

37. What is the difference between photons and electrons in radiotherapy?

38. What is palliative radiotherapy?

39. *What is curative radiotherapy?*

40. *What is neoadjuvant radiotherapy?*

41. *What is adjuvant radiotherapy?*

42. *What is hypofractionation?*

43. *What is hyperfractionation?*

44. *What is boost dose in radiotherapy?*

45. *What is immobilization in radiotherapy?*

46. *What is organ at risk (OAR)?*

47. *What is treatment field?*

48. *What is isocenter in radiotherapy?*

49. *What is tumor localization?*

50. *What is radiation oncology?*

51. *What is the role of a medical physicist in radiotherapy?*

52. *What is the role of a radiation therapist?*

53. *What is the role of a radiation oncologist?*

54. *What is a treatment course?*

55. *What is dose homogeneity?*

56. *What is dose conformity?*

57. *What is beam energy in radiotherapy?*

58. *What is radiobiology?*

59. *What is tumor control probability (TCP)?*

60. *What is normal tissue complication probability (NTCP)?*